

Q1

$$\text{Dataword} = 11011011$$

$$\text{Divisor} = x^3 + 1 = 1001$$

$$\therefore \text{Dividend/Augmented Dataword} = 11011011000$$

$$\begin{array}{r} 1001 \overline{) 11011011000} \\ \underline{1001} \\ 1001 \\ \underline{1001} \\ 001100 \\ \underline{1001} \\ 1010 \\ \underline{1001} \\ 011 \end{array}$$

$$\therefore \text{Remainder} = 011$$

$$\therefore \text{Codeword} = 11011011011$$

Corrupting Left most third bit 11111011011

$$\begin{array}{r} 1001 \overline{) 11111011011} \\ \underline{1001} \\ 1101 \\ \underline{1001} \\ 1000 \\ \underline{1001} \\ 1110 \\ \underline{1001} \\ 111 \\ \underline{1001} \\ 110 \\ \underline{1001} \\ 100 \neq 0 \end{array}$$

$$\therefore \text{Remainder} \neq 0$$

$\therefore$ Error Detected

Q₂

$$\begin{array}{r} \text{(I)} \quad 3046 \\ \quad \quad ABDC \\ \quad \quad 02B5 \\ \quad \quad \underline{0E30} \\ \quad \quad ED07 \end{array}$$

$$\begin{aligned} \therefore \text{CHECKSUM} &= \text{NOT}(ED07) \\ &= 12F8 \quad (\text{Ans}) \end{aligned}$$

(II)

$$\begin{array}{r} 3047 \\ \quad \quad ABDC \\ \quad \quad 02B4 \\ \quad \quad \underline{0E30} \\ \quad \quad ED07 \end{array}$$

changed.

$$\begin{aligned} \therefore \text{CHECKSUM} &= \text{NOT}(ED07) \\ &= 12F8 \end{aligned}$$

$$\therefore \text{Sender's Checksum} = \text{Receiver's Checksum}$$

$\therefore$ No ERROR Detected

(III)

The Data's are changing in such a way that the checksum remained the same, authorizing the receiver that "Bro, let me go, Everything is OK. Cool Down". Even though the data become corrupted, the receiver fails to detect error.

Q₃

$$d(00000, 01011) = 3$$

$$d(01011, 10101) = 4$$

$$d(10101, 11110) = 3$$

$$d(00000, 10101) = 3$$

$$d(00000, 11110) = 4$$

$$d(01011, 11110) = 3$$

$$\therefore d_{\min} = s + 1 \quad ; \quad s = \text{error}$$

$$\Rightarrow s = 3 - 1;$$

= 2 bit errors can be successfully detected.

$$\therefore d_{\min} = 3$$

$$d_{\min} = 2 + 1$$

$$\Rightarrow t = \frac{d_{\min} - 1}{2} = \frac{3 - 1}{2} = 1 \text{ bit corrected possible.}$$

Q4

(i) $n = \text{bits of codeword} = 5$

$k = n \quad n \text{ dataword} = 2$

$\therefore C(n, k) = C(5, 2)$

(ii)

$$\begin{array}{r} 01011 \\ \oplus 10101 \\ \hline 11110 \end{array}$$

(iii)

$d(00000, 01011) = 3$

$d(00000, 10101) = 3$

$d(00000, 11110) = 4$

$d(01011, 10101) = 4$

$d(01011, 11110) = 3$

$d(10101, 11110) = 3$

$\therefore d_{\min} = 3$

$\therefore d_{\min} = 3 + 1 \Rightarrow s = 3 - 1 = 2$ ^{bit of} errors can be detected

$\therefore d_{\min} = 2t + 1 \Rightarrow t = \frac{3-1}{2} = 1$ bit of error can be corrected.

Q5

$$\text{Dataword} = x^6 + x^4 + x^3 + x + 1$$

$$\text{Divisor} = x^4 + x^2 + x + 1$$

$$\therefore \text{Augmented Dataword} = x^4 x (x^6 + x^4 + x^3 + x + 1) = x^{10} + x^8 + x^7 + x^5 + x^4$$

$$\begin{array}{r}
 x^4 + x^2 + x + 1 \overline{) x^{10} + x^8 + x^7 + x^5 + x^4} \quad (x^6 + x^2 + x) \\
 \underline{x^{10} + x^8 + x^7 + x^6} \\
 x^6 + x^5 + x^4 \\
 \underline{x^6 + x^4 + x^3 + x^2} \\
 x^5 + x^3 + x^2 \\
 \underline{x^5 + x^3 + x^2 + x} \\
 x
 \end{array}$$

$$\therefore \text{Remainder} = x$$

$$\begin{aligned}
 \therefore \text{Codeword} &= \text{Augmented Dataword} + \text{remainder} \\
 &= x^{10} + x^8 + x^7 + x^5 + x^4 + x
 \end{aligned}$$

Q6

Sender

8

12

14

0

9

$$(43)_b = (101011)_2$$

101011 $\approx$ 6 bit

Doing wrap summing carry bit,

$$\begin{array}{r}
 01011 \\
 + 1 \\
 \hline
 01100
 \end{array}$$

Doing 1's complement,

$$(10011)_2 = 19$$

Sender's checksum = Receiver Checksum

Error Not Detected, checksum arithmetic fails.

Q7

① $d(0001110, 0110110) = 3$

$d(0001110, 1010101) = 5$

$d(0001110, 1100001) = 6$

$d(0110110, 1010101) = 4$

$d(0110110, 1100001) = 5$

$d(1010101, 1100001) = 3$

②

$d_{\min} = 3$

$s = d_{\min} - 1 = 2$

$t = \frac{d_{\min} - 1}{2} = 1$

③

codeword ①

0 1 1 0 0 1 0

⇒ can fix the error as error # bits is 1

codeword ②

1 0 1 1 1 0 → 6 bit

⇒ Receiver has 7 bit. So, it sends the codeword back.

Q8

②

4507

01A4

A309

0B00

F4B4

∴ CHECKSUM = NOT(F4B4)
= 0B4B

③

4508

01A3

A309

0B00

F4B4

∴ CHECKSUM = NOT(F4B4)
= 0B4B

Written Before, Reasons of
failure

Q10

$$\text{Dataword} = 100110$$

$$\text{Remainder} = 100 \rightarrow \text{length} = 3 \rightarrow \text{Divisor length} = 4$$

$$\therefore \text{Divisor} = 1100$$

$$\begin{array}{r}
 1100 \overline{) 100110100} \quad (111011 \\
 \underline{1100} \\
 1011 \\
 \underline{1100} \\
 1110 \\
 \underline{1100} \\
 1010 \\
 \underline{1100} \\
 1100 \\
 \underline{1100} \\
 0
 \end{array}$$

$\therefore$ NO ERROR DETECTED. 😊

Q8

$$(i) x^6 + x^5 + x = \text{codeword} = 1100010$$

$$\text{Quotient} = x^3 + x^2 + x = 1110$$

$$C(7, 4); n=7, k=4; r=7-4=3$$

$$\therefore \boxed{1100} \boxed{010} \rightarrow \text{Redundant Bit/Remainder}$$

$\hookrightarrow$ Dataword.

$$\therefore \text{Augmented Dataword/Dividend} = 1100000$$

$$\therefore \text{Dividend} = (\text{Divisor} * \text{Quotient}) \oplus \text{Remainder}$$

$$\Rightarrow 1100000 = (x * 1110) \oplus 010$$

$$\Rightarrow x * 1110 = 1100010$$

$$\begin{array}{r}
 1100000 \\
 0000010 \\
 \hline
 1100010
 \end{array}$$

$$\begin{array}{l}
 \text{Dataword} = \\
 x^6 + x^5 \\
 \text{Remainder} = x
 \end{array}$$

$$\begin{array}{r}
 1110 \) \ 1100010 \ (\ 1011 \\
 \underline{1110} \\
 1001 \\
 \underline{1110} \\
 1110 \\
 \underline{1110} \\
 0
 \end{array}$$

$$\therefore x = 1011$$

$$\therefore \text{Divisor} = 1011 = x^3 + x + 1$$